

Proof. 1. Edge-case verification. If $\mathbf{b} = \mathbf{0}$, then $S_{b^2} = 0$, whence $S_{ab} = 0$ and the right-hand side of the Cauchy–Schwarz inequality equals $S_{a^2} \cdot 0 = 0$, while the left-hand side is $(0)^2 = 0$. Hence the inequality holds with equality. The same argument applies when $\mathbf{a} = \mathbf{0}$. Consequently we may assume $S_{b^2} > 0$ (the case $S_{a^2} = 0$ is analogous).

2. Construction of a non-negative quadratic. For $t \in \mathbb{R}$ define

$$Q(t) = \sum_{i=1}^n (a_i - t b_i)^2.$$

Each summand is a square, so $Q(t) \geq 0$ for every t . Expanding each term,

$$(a_i - t b_i)^2 = a_i^2 - 2t a_i b_i + t^2 b_i^2,$$

we obtain

$$\begin{aligned} Q(t) &= \sum_{i=1}^n (a_i^2 - 2t a_i b_i + t^2 b_i^2) \\ &= S_{a^2} - 2t S_{ab} + t^2 S_{b^2}. \end{aligned}$$

Thus $Q(t)$ is a quadratic polynomial

$$Q(t) = S_{b^2} t^2 - 2S_{ab} t + S_{a^2}.$$

3. Discriminant criterion. A quadratic $At^2 + Bt + C$ with $A > 0$ that satisfies $At^2 + Bt + C \geq 0$ for all $t \in \mathbb{R}$ must have non-positive discriminant,

$$\Delta = B^2 - 4AC \leq 0.$$

Here $A = S_{b^2}$, $B = -2S_{ab}$ and $C = S_{a^2}$, so

$$\Delta = (-2S_{ab})^2 - 4S_{b^2}S_{a^2} = 4(S_{ab}^2 - S_{a^2}S_{b^2}).$$

Since $Q(t) \geq 0$ for all t , we have $\Delta \leq 0$, i.e.

$$S_{ab}^2 \leq S_{a^2} S_{b^2}.$$

Writing the sums explicitly yields the Cauchy–Schwarz inequality

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right).$$

4. Equality condition. Equality occurs exactly when $\Delta = 0$, i.e. when $S_{ab}^2 = S_{a^2}S_{b^2}$. In this case the quadratic $Q(t)$ has a double root t_0 and can be written as

$$Q(t) = S_{b^2} (t - t_0)^2 \geq 0, \quad Q(t_0) = 0.$$

Because $Q(t_0) = \sum_{i=1}^n (a_i - t_0 b_i)^2 = 0$, each summand must vanish: $a_i = t_0 b_i$ for all i . Hence $\mathbf{a} = t_0 \mathbf{b}$, i.e. the two vectors are linearly dependent (including the trivial case where one vector is zero). Conversely, if $\mathbf{a} = t_0 \mathbf{b}$ for some $t_0 \in \mathbb{R}$, then every term in $Q(t)$ vanishes at $t = t_0$, so the inequality becomes an equality.

5. Conclusion. The non-negativity of the quadratic $Q(t)$ together with the discriminant condition yields

$$\boxed{\left(\sum_{i=1}^n a_i b_i\right)^2 \leq \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right)}$$

for all real numbers $a_1, \dots, a_n$ and $b_1, \dots, b_n$. Equality holds precisely when $\mathbf{a}$ and $\mathbf{b}$ are linearly dependent (including the case that one of them is the zero vector). □